Four ingredients

A sidewalk is concrete: gravel, sand, gray cement powder and water mixed together. Cement is only one ingredient, not the whole thing.

Paste that glues

Cement and water make a gray paste. It coats every stone and grain of sand and, once hard, holds them all tightly together.

Pour, spread, wait

Wet concrete is poured into a wooden form and smoothed. Cement and water then react and link together, so it hardens, even under water.

Curing takes days

A new slab is kept damp for about a week while the reaction goes on. This is curing. Most strength builds in the first month.

A line for cracks

Concrete shrinks as it hardens and can crack. Grooves make thin lines in the slab, so cracks tend to follow them in neat, straight lines.

Strong when squeezed

Concrete is strong when squeezed but weaker when bent. Packed gravel under the slab helps spread the weight so it bends less.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

How a Sidewalk Is Made and How It Works